Muka Lingam

AI SOLUTION ARCHITECT

IICL · Hyderabad, India

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CORE EXPERTISE

Agentic AI Systems	97%
LLM Integration & RAG	96%
LangChain / LlamaIndex	93%
CrewAI / Google ADK	90%
AWS & GCP Cloud	88%
Prompt Engineering / MCP	95%
Python · FastAPI	94%
Voice AI & Conversational	87%

TECH STACK

GPT-4o · Claude 3.5 · Gemini 2.0 · Llama 3.1 · CrewAI · Google ADK · LangGraph · AutoGen · MCP · pgvector · Pinecone · PostgreSQL · AWS Lambda · SageMaker · EC2 · ECS · GCP Vertex AI · Docker · Kubernetes · FastAPI · React · PyTorch · YOLO · ElevenLabs · Mendix

EDUCATION & CERTIFICATIONS

B.Tech · Mechanical Engineering
St. Martin's Engineering College · JNTUH
2015 – 2019 · 70%

AI & Machine Learning
Universiti Teknologi Malaysia (UTM)
🏅 Certified · AI / ML Programme

AGENT FRAMEWORK PROFICIENCY

LC=LangChain LI=LlamaIndex LG=LangGraph

Principal-level AI Solution Architect with **7+ years** delivering production-grade agentic AI systems across FinTech, Healthcare, Government & AdTech. Architect of multi-agent orchestration platforms (CrewAI, Google ADK, LangGraph, AutoGen), RAG pipelines with vector stores, and Voice AI systems. Hands-on cloud deployment expertise on **AWS** (Lambda, SageMaker, ECS, EC2) and **GCP** (Vertex AI, Cloud Run). Track record translating complex AI architectures into regulated, enterprise-grade production deployments.

AGENTIC AI ARCHITECT · RAG & VECTOR DBS · LANGCHAIN · LLAMAINDEX · CREWAI · AWS & GCP PRODUCTION · GOOGLE ADK · LANGGRAPH · 7+ YRS · 1M$+ IMPACT

EXPERIENCE

AI Solution Architect
Jan 2026 – Present
IICL (Intelligence India.Com Limited) · Hyderabad · On-site

- Architecting Nexlayer AI — domain-specific multi-agent platform (CrewAI + Google ADK) for Finance, Government & Healthcare with India-first compliance (DPDP, RBI, IRDAI).
- Designed Fixly AI (AI-native ITSM) & Vonex AI (enterprise voice agent SaaS); deployed full pipelines on AWS ECS & GCP Vertex AI; cost-optimised Kubernetes saving $15K+.

Lead AI Engineer
Apr 2025 – Jan 2026 · 10 mos
Lowcode Labs · Hyderabad · On-site

- Deployed enterprise agentic ecosystems (GPT-4o, Claude, Gemini) with RAG via pgvector, serving 500+ users at 99.5% uptime; cut deployment time 60% via MLOps automation.
- Built real-time Voice AI agents (ElevenLabs + WebSocket), WhatsApp AI agents in Mendix & MedGemma 4B healthcare AI on AWS EC2; reduced cloud costs by $15,000+.

Lead AI Engineer
Nov 2023 – Apr 2025 · 1 yr 6 mos
The Audience Street Inc · Hyderabad · On-site

- Led Ninadata AdTech platform: LangChain-powered ad intelligence with intelligent model routing (DALL-E, GPT-4, Claude) — 20% engagement uplift, 40% content quality gain.

Senior Data Scientist & Data Scientist
Dec 2021 – Feb 2024 · 2 yrs 3 mos
Spotflock · Hyderabad · Full-time

- INCOIS ocean prediction (Transformers + LSTM): 85% accuracy, 25-30% analytics time saved — officially recognised by INCOIS/Ministry of Earth Sciences.
- Sembcorp Gauge Monitoring (CV + YOLO): 900+ labour hours saved, 47% efficiency gain, **US$1M+ operational savings**.

Associate Data Scientist & Intern
Feb 2020 – Dec 2021 · 1 yr 11 mos
INNODATATICS · Hyderabad · Full-time

- Developed predictive maintenance models (Python, scikit-learn) achieving 30% reduction in equipment downtime; Haryana Govt. electricity load forecasting (ARIMA + LSTM + Prophet).

Design Engineer
Jul 2019 – Jan 2020 · 7 mos
United Systems & Projects India Pvt Ltd · Hyderabad · On-site

- Mechanical design & production engineering; early foundation in structured problem-solving that transitioned into AI/ML career.

AI PRODUCTS & KEY PROJECTS

AI Universee — Founder
Unified multi-modal AI platform routing across GPT-4o, Claude, Gemini, DALL-E, Midjourney & Runway ML via one interface with domain-specific templates.

GPT-4o · Claude · Gemini · DALL-E · FastAPI · React

TubeQuiz AI — Founder
RAG-powered platform converting video content into adaptive quizzes via multi-modal analysis (vision + audio + text extraction) with pgvector.

RAG · GPT-4o · pgvector · CV · React

INCOIS Ocean Prediction — Gov · MOES
Transformers-based ocean current forecasting for Indian National Centre for Ocean Information Services. 85% accuracy; 25-30% analytics time saved. Officially recognised.

▲ 85% accuracy · Recognised by INCOIS/MOES

Transformers · LSTM · PyTorch · NumPy

HERC – Healthcare Emergency — Healthcare AI
AI-powered Healthcare Emergency Response & Coordination system — multi-agent CrewAI workflow for triage, resource allocation & real-time clinical decision support.

CrewAI · Claude API · FastAPI · PostgreSQL · AWS

Nexlayer AI — Production
Domain-specific agentic platform for Finance, Government & Healthcare — CrewAI + Google ADK orchestration, DPDP/RBI compliance, India-first architecture.

CrewAI · Google ADK · LangGraph · AWS · GCP

Sembcorp Gauge Monitor — $1M+ Saved
Edge computer vision system (YOLO + OpenCV) for real-time industrial gauge monitoring & predictive maintenance at Sembcorp manufacturing plants.

▲ 900+ hrs saved · 47% efficiency · US$1M+

YOLO · OpenCV · PyTorch · AWS EC2

Vonex AI / VoxaHub
Enterprise voice agent SaaS — ElevenLabs + Claude API, real-time WebSocket streaming, branded persona management & sentiment-aware conversation flows.

ElevenLabs · Claude API · WebSocket · FastAPI · React

MediConnect – Healthcare AI
MedGemma 4B RAG integration in Mendix — medical literature KB, drug databases, clinical guidelines with HIPAA-conscious audit logging on AWS EC2.

MedGemma 4B · RAG · Mendix · AWS EC2 · SSL

THOUGHT LEADERSHIP & COMMUNITY

- **East Coast Maritime & Logistics Summit 2025** — Invited speaker presenting AI innovations to industry leaders & Hon. Chief Minister of Andhra Pradesh.
- **Medium Writer** — Published technical deep-dives on LLMs, RAG pipelines, MCP, agentic AI & enterprise AI strategy at medium.com/@utlamuka.
- **Boston Institute of Analytics** — Empanelled trainer for Data Science & AI, Generative AI & Agentic AI Development modules.
- **Portfolio & Open Source** — Live projects, demos & repositories at mukalingam.github.io & github.com/Mukalingam.